

Program Question_2a

```
Implicit none
real, Dimension(5)::x,y
real::x_interp,y_interp
Integer::i,j
real::term
Open(Unit=15,file='a3q2a_in.txt')
Open(Unit=17,file='a3q2a_out.txt')
!Data point Initializaation
!x=[1.0,2.0,4.0,5.0,7.0]
!y=[52.0,5.0,-5.0,-40.0,10.0]
Write(17,*)"Enter the value of x"
Do i=1,5
    Read(15,*)x(i)

End Do
Write(17,*)"Enter the value of y"
Do j=1,5
    Read(15,*)y(j)
end do

!Interpolation points
x_interp=3.0
y_interp=0.0
!Lagrange Interpolaiion formula
Do i=1,5
    term=y(i)
    Do j=1,5
        If(i/=j)then
            term = term*((x_interp-x(j))/(x(i)-x(j)))
        End If
    End Do
    y_interp=y_interp+term
End Do
Write(17,*)"The Interpolation x=3",y_interp
End Program
```